

Online Retail Sales Database

Objective

The objective of this project is to design and implement an Online Retail database system that manages customers, products, and orders. The system focuses on maintaining structured data and handling relationships between different entities using relational database concepts.

System Overview

The Online Retail system allows customers to purchase products and place orders. Each order can contain multiple products, and customer order history can be tracked. The system ensures proper data organization and efficient data retrieval using SQL.

Tables

Customer

Stores customer details.

Attributes: customer_id, name, email, phone

Product

Stores product information.

Attributes: product_id, product_name, price, stock

Orders

Stores order details.

Attributes: order_id, customer_id, order_date

Order_Items

Stores products included in each order.

Attributes: order_item_id, order_id, product_id, quantity

ER Diagram

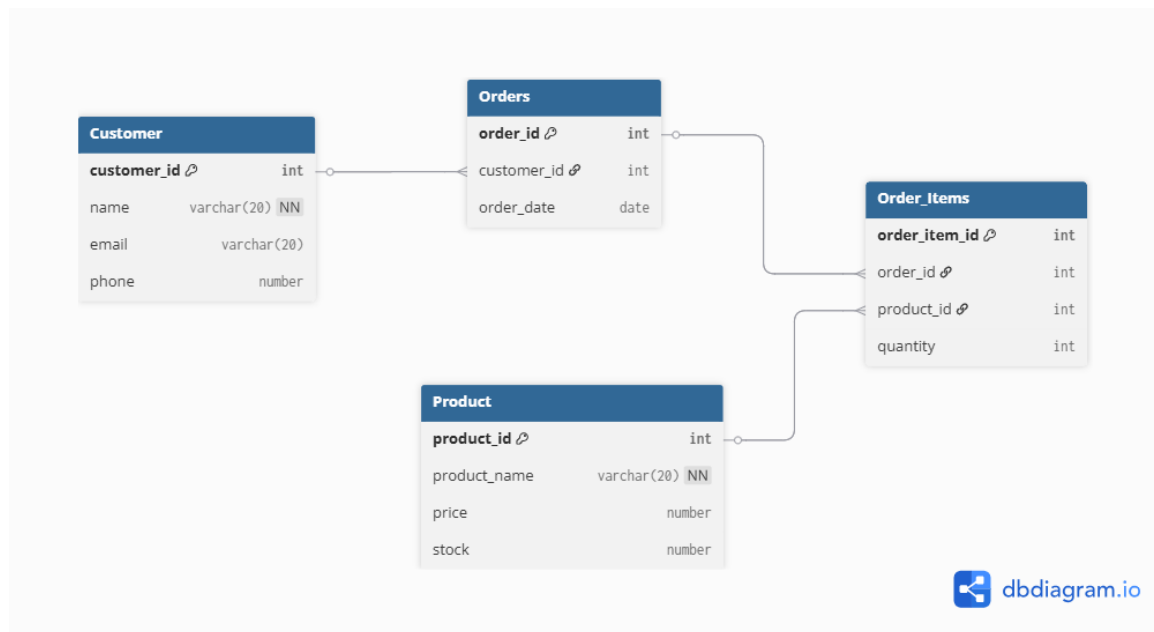


Figure: ER Diagram of Online Retail Management Database

Relationships

1. Customer → Orders (1:M): One customer can place multiple orders
2. Orders → Order_Items (1:M): One order can contain multiple items
3. Product → Order_Items (1:M): One product can be part of multiple orders
4. Orders ↔ Product (M:M via Order_Items): Managed using Order_Items table

Key Features

Maintains structured data using relational database design.
Applies normalization to reduce redundancy.
Handles many-to-many relationships using Order_Items table.
Tracks customer orders and product details.
Ensures data integrity using primary key and foreign key constraint.

Technologies Used

Oracle Database 11g, SQL, SQL*Plus, dbdiagram.io

Conclusion

This project demonstrates a structured database design for an online retail system. It helps manage customers, products, and orders efficiently using SQL.